

Continue “ Project Status Snapshot

Revision: 24 Last modified: 2026-08-18T21:10:20Z

TERMINAL STATE (this write) “ Â§11.4.126 endless-loop terminal condition MET

- **Pre-build: 27 passed / 0 failed** “ all invariants GREEN including new CM-RESOURCE-PRESSURE-SIGNATURE-CHECK (invariant 25) and post-fix CM-DOCS-CHAIN-ENGINE-VERIFY (invariant 24)
- **3 boba remotes IN SYNC at fab6707a**
- **Â§11.4.209 review:** 0 BLOCKING / 0 IMPORTANT open / 5/6 MINOR fixed / 1 MINOR honest SKIP / 2 NIT non-blocking / Fable-xhigh re-review owed as Â§11.4.197 upgrade
- **Â§11.4.115(F) polarity:** 5/5 RED CONFIRMED across real pathological fixtures
- **Resource-pressure timer:** LIVE + ARMED (next fire 22:42 CEST)
- **1 subagent dispatched** for Â§11.4.108 rebuild + runtime-signature verification

Session commit summary (~20 boba commits + 4+ constitution commits since 457cca4)

SESSION 2026-08-15 â†’ 2026-08-18 (Session 4db6eadb-03e7)

Working state at last write: 3 boba remotes IN SYNC at 9c282ff. resource-pressure-signature-check systemd-user timer LIVE (next fire 22:42 CEST). 3 subagents in flight: Task #78 (IMPORTANT-1 real per-signature RED fixtures), Task #79 (IMPORTANT-2 SQLite differential-dump discipline), Task #80 (docs_chain codegraph-status sync/verify contradiction).

Major deliverables this session (in commit order)

- **Amendment round v68 landed** (Constitution + 4 mirrors + docs_chain + boba pointer, 22 new anchors Â§11.4.239-262 + 5 extensions + Â§11.4.140/141 collision resolution â†’ Â§11.4.255/256 re-mint)
- **Tag helixconstitution-v68** published on 4 platform releases (GitHub Ã— 2 + GitLab Ã— 2) + 8 constitution git remotes
- **BOB-064:** durable-run.sh helper (commit 20571a0)
- **BOB-066:** audit L1/L2/L4 wired (commit b9c2c50)
- **BOB-067:** Jackett cookie-login + HelixQA fake (commits 6231200 + 6dfc756)

- **Loader auto-discovery** + Jackett submodule bump v0.24.2353 â†’ v0.24.2406
- **All 5 tracker credentials wired** (RUTRACKER, NNMCLUB, RUTOR, KINOZAL, IPTORRENTS) with Â§11.4.10.A pre-store leak audits
- **cookies.txt convention** at ~/Downloads with atomic auto-load into .env chmod 600
- **systemd integration** surviving reboot (boba.target + boba-stack.service + boba-webui-bridge.service, linger=yes)
- **Production install.sh** authored + LAN access verified on 192.168.1.90:{7185,7186,7187,7188,7189,9117}
- **BOB-112 boba-jackett /healthz DDoS mitigation:** sync.RWMutex TTL cache, live wrk **97.1% â†’ 0.0% timeout** (400/412 â†’ 0/812,149 requests) â€” commit e2a2e3e
- **BOB-113:** install-dev-tools.sh (wrk/hey/siege) â€” commit c7dfdde
- **Task #66 commit-push-all.sh â€”“scope flag** (BOB-068 sweep-pattern interim fix) â€” commit 0972cbc
- **BOB-108:** workable-items export Revision-counter regression fix â€” constitution 4a17867 + boba 855ce65 + 3520621
- **CodeGraph 1.5.0 nested-.gitignore regression:** upstream issue #1567 + draft PR #1568 at colbymchenry/codegraph
- **Task #72 docs_chain unpushable-gitlink:** cherry-pick + push resolution (docs_chain 8ccf505 + constitution 774ac57 + boba a4c6c7a + 41da88a)

CRITICAL 2nd forced-logout incident (2026-08-18 20:50:59)

- CONST-033 triage clean: no kernel OOM, no systemd-oomd trigger, no lid-suspend, no forbidden mechanism
- Â§12.12 EAGAIN cascade signature at 20:45:48 confirmed (jackett SocketException (11) to iptorrents+kinozal+rutracker simultaneously)
- 15 GB pathological ugrep reaped post-relogin: ugrep -o with . \{0,120\} variable-length context + 3-way alternation on 14K-line CLAUDE.md â€” freed 16 GB instantly, PSI Avg10 1.77 â†’ 0.08
- Â§11.4.6 UNCONFIRMED: exact SIGKILL source unattributable from systemd journal â€” PENDING_FORENSICS (task #79)
- **Fix commit 1f42357:** CONST-033 challenge false-positive fix + NEW resource_pressure_signature_challenge.sh (5 signatures, Â§11.4.115 polarity) + docs/incidents/2026-08-18-perceived-forced-logout-2nd.md + 8 evidence artifacts
- **Â§11.4.238 ledger entry 98412bf â€” FORCED-LOGOUT-2026-08-18-2ND** (Rev 6â†’7, count 14â†’15)
- **Task #77 wire commit ecb3bfe:** pre_build invariant 25 CM-RESOURCE-PRESSURE-SIGNATURE-CHECK + hourly systemd-user timer INSTALLED + ARMED (next fire 22:42 CEST)
- **Persistent memory playbook** at ~/.claude-claude4/â€¦/memory/forced_logout_incidents.md
- **Task #41 BOB-068 investigation a7e55f9:** verdict = NO DAEMON. Root cause = shared-checkout race on default git add -A + shared docs/workable_items.db. Â§11.4.179 clone-isolation (Task #67 proposal) confirmed correct architectural fix.

Â§11.4.209 review verdict: NO-GO with 3 IMPORTANT + 6 MINOR + 2 NIT

- **IMPORTANT-1** (BOB-116 polarity claim threshold-only, not real-signature “ initially referenced as BOB-076 informal label, corrected 2026-08-18; BOB-076 is a distinct unrelated Type=Task DB item, see docs/incidents/2026-08-18-perceived-forced-logout-2nd.md): fix subagent in flight (task #78)
- **IMPORTANT-2** (3520621 DB commit missing differential SQLite dump): fix subagent in flight (task #79)
- **IMPORTANT-3** (HTTPS git URL in install-dev-tools.sh): FIXED inline commit 9c282ff (both real invocation + advisory message ‘ git@)
- MINOR findings tracked as followups
- Â§11.4.209 Fable-xhigh re-review OWED (this review ran on opus fallback per Agent-tool effort=? Â§11.4.231(F)(b) honest boundary)

To resume

- 3 subagents in flight: notifications will arrive
- New commits will push automatically (their briefs specify commit + push)
- Task tracker #40, #42 remain deferred (need operator design decisions)
- Constitution submodule pointer: constitution at 4a17867 (BOB-108 fix) ‘ boba 9c282ff ‘ docs_chain 8ccf505
- Everything else in Session 15 CONTINUATION section below remains valid history **Session:** 2026-08-10/11 (Session 16 “ HUGE session: pre_build brought to 25/25 GREEN with six new invariants (INV0/19/20/21/22/23), Â§11.4.238 QA-discovery gates wired, Â§11.4.84 subagent-mutation fence Layers 1+3 landed, workable-items diff false-positive fixed + new correct-history- evidence subcommand, 39 BOB backfills closing a Â§11.4.197 mass violation, three stress+chaos suites landed GREEN, HelixQA banks parametrized for RD2-34, QA_DISCOVERY_LEDGER at Rev 3 with a new automated-background-scan channel. Definitive host-kill root cause identified “ ALT Linux KillUserProcesses=true compile-time default; mitigation config staged at scratchpad/20-kill-exclude-milosvasic.conf but STILL BLOCKED on operator sudo.) **Last commit (boba main):** 1d0268f (chore(hygiene): gitignore scratchpad/ + update helixqa submodule URL to renamed upstream). Prior big session-16 commit: 4b7b21d (session-16: pre_build 25/25 GREEN “ Â§11.4.238 gates + Â§11.4.84 fence + backfills + stress+chaos + BOB-009/010 fix). **Branch:** main **Working tree:** CLEAN (post-session-16 commits landed + pushed). Only in-flight edit: this CONTINUATION.md revision itself.

Send continue to pick up exactly where we left off. This file is the single source of truth for session handoff (Â§12.10 / Â§11.4.131).

CURRENT STATE â€” Session 16 (2026-08-10/11): pre_build 25/25 GREEN + Â§11.4.238/Â§11.4.84 hardening + 39 BOB backfills + stress+chaos + BOB-009/010 fix

Operator asked for the Â§11.4.238 QA-discovery regime to actually bind (per its zero-grace-period â€œno escape hatchâ€” clause) and for the Â§11.4.197 mass-violation from Session 15â€™s audit backlog to be closed. Delivered as one big commit (4b7b21d, 37 files) + a hygiene follow-up (1d0268f), both pushed to all upstreams on boba main. Constitution + HelixQA submodule pointers advanced + pushed. This session survived multiple kill events (below) â€” the constitution submodule and HelixQA submodule work landed BEFORE the boba-side work, so no in-flight cross-repo saga was left half-applied (Â§11.4.191 preserved).

What landed (Session 16 â€” all published)

boba main-repo (4b7b21d + 1d0268f): - **Pre-build 25/25 GREEN** â€” six NEW invariants in scripts/pre_build_verification.sh + scripts/pre_code_review.sh: **INV0** CM-PREFLIGHT-INTERPRETER (catches bare python3 -m pytest vs .venv/bin/python ABI drift at collection time â€” RD2-41b/RD2-43); **INV19-21** Â§11.4.238 (ledger fresh + counts aligned, every out-of-band entry carries escape-audit: + new-check:, propagation literal in the Â§11.4.157 mirror set â€” INV21 wrong-seam bug corrected to check constitution/{CLAUDE,AGENTS,QWEN,GEMINI}.md, not project root); **INV22** CM-DOCS-CHAIN-STEP1-REAL-INVOCATION (Â§11.4.238 RD2-41a retroactive catcher, anchored ^\s*ERROR: grep, structure-not-substring per Â§11.4.201(7)(a)); **INV23** CM-NO-PRODUCTION-MUTATION-RESIDUE (Â§11.4.84 layer 3, string-concat de-mutation so scanner doesnâ€™t self-match, FIXTURE_ROOT env override for golden-good/golden-bad). pre_code_review.sh: mutation-marker carrier false-positive fixed 36â†’0 hits. - **Â§11.4.84 subagent-mutation fence** (crypto.go RED-window incident from Session 15/16): **Layer 1** â€” subagent briefing template with 3 mandatory isolation patterns (git worktree / atomic mutate-run-restore / mutation tool); **Layer 3** â€” INV23 above. **Layer 2** (post-tool-use hook) DEFERRED pending runtime capability check; design at scratchpad/task-20-84-fence-design.md. - **Â§11.4.238 QA discovery ledger** (docs/QA_DISCOVERY_LEDGER.md Rev 3): NEW **INC-2026-08-10** entry â€” crypto.go GCM auth-bypass mutation left in production file during Agent Hâ€™s RED window, caught by background security-review plugin (a new discovery channel: automated_background_scan). Full escape-audit + new-check documented. Table split 7 â†’ 8 out-of-band. - **workable_items.db reconciled + expanded** (63 â†’ 102 items): **39 BOB backfills** for ~40 RD2-/GA-/RW-* audit-doc items that had no tracker rows (Session 15â€™s Â§11.4.197 mass violation â€” CLOSED). BOB-064..067 for Lava P1..P4 (RD2-15 reservation); BOB-068..102 for remaining audit items. **BOB-009 + BOB-010** evidence_path corrections via a NEW workable-items correct-history-evidence subcommand (constitution submodule) â€” no more raw SQL, no more falsely-implied reopen. workable-items diff false-positive

fixed (bare --db invocations were flagging every row as absent-from-MD).
 validate: OK " 102 items. - **Stress+chaos tests (Â§11.4.85)**: NEW
 tests/stress/test_tracker_fetch_stress_chaos.py (9/9 GREEN, 27.5s,
 9 JSON evidence files under qa-results/stress_chaos/); NEW
 qBitTorrent-go/tests/integration/
 scheduler_hooks_sse_stress_chaos_test.go (6/6 GREEN + -race clean,
 5.6s); +702 lines in qBitTorrent-go/tests/integration/
 jackett_db_test.go " 5 new chaos cases (byte-corruption, concurrent-
 writer, WAL-sidecar, master-key-rotation, mid-tx SIGKILL), 5.8s -race clean. -
Hygiene follow-up (1d0268f): .gitignore scratchpad/ (24 files/176K
 session artifacts, Â§11.4.10 scan clean); .gitmodules helixqa URL updated
 HelixQA.git " qa.git following upstream repo rename (GitHub warned
 "This repository moved"💎), git submodule sync propagated.

constitution submodule (pushed to all 8 remotes): 4a2adac (fix workable-
 items diff false-positive + new correct-history-evidence subcommand +
 sudo/su quote-aware guard) + 03f0b0c (merge remote 6f960ca design-
 toolkit pointer bump per Â§11.4.188).

HelixQA submodule (4289cf4, pushed): banks(boba) " added BOBA-
 PRX-009/010 RD2-22 Challenges + parametrized /Volumes/T7 hardcoded
 paths (RD2-34 closed).

Session-kill root cause " DEFINITELY IDENTIFIED, mitigation staged, BLOCKED on sudo

- **Root cause** (per Session 15's live busctl finding): ALT Linux ships
 systemd-logind with the **compile-time default**
KillUserProcesses=true " every GDM/GNOME session close
 SIGKILLS the entire user@1000.service slice (tmux, Claude Code,
 everything) regardless of Linger=yes. Confirmed multiple hits again
 this session.
- **Mitigation staged**: scratchpad/20-kill-exclude-milosvasic.conf
 (a logind.conf.d drop-in for KillExcludeUsers=milosvasic "
 narrower than blanket KillUserProcesses=no). Â§11.4.6 HONEST
 GAP: the file was described in the session hand-off brief but is NOT
 visible in the scratchpad listing at write time " either it was staged
 elsewhere, wiped by a kill event, or the brief refers to a design not yet
 materialized on disk; next session MUST re-verify + author the drop-in
 fresh if absent.
- **Blocked on**: sudo cp of the drop-in into /etc/systemd/
 logind.conf.d/ + systemctl restart systemd-logind. Per repo
 CLAUDE.md Â§ "Host Power Management " Hard Ban💎 AND
 the constitution's Â§11.4.109 PreToolUse guard, the agent's
 sudo is CORRECTLY refused; the operator must run the copy + restart
 themselves (the !-prefix-in-permissions path is the sanctioned operator-
 executed channel). Until then, every session remains at risk of the same
 abrupt kill.

Pending operator Â§11.4.66 decisions

1. **Apply the KillUserProcesses fix** (narrower KillExcludeUsers=milosvasic vs blanket KillUserProcesses=no) “ see above. HIGHEST priority: every future session is at risk until this lands.
2. **Â§11.4.84 Layer 2 post-tool-use hook** “ runtime capability check owed; design at scratchpad/task-20-84-fence-design.md.
3. **RD2-22 live-verify still owed** (source fix done Session 15; the full tests/contract/+tests/unit/ regression sweep was killed mid-run by the logind issue and still needs to complete + a live curl-verify 401/200 on a running container + the Â§11.4.135 regression guard + HelixQA Challenge entry).
4. **Â§11.4.234 dedicated commit/push wrapper** (task #25) “ deferred with operator go-ahead; direct commits used this session.

In-flight work carried forward

- **pytest sweep v5** “ the RD2-22 regression sweep still owed (pending item 3 above).
- **Task #27 (suite-scope repro)** “ still needs a scope-widened reproduction pass. Details in the operator brief that opened this session; re-derive from docs/workable_items.db + Session-15 audit if the working note is not in scratchpad.

What to do FIRST on next-session pickup

1. **Land the logind fix** (mitigation config “ /etc/systemd/ logind.conf.d/ + systemctl restart systemd-logind). Verify with busctl get-property org.freedesktop.login1 /org/freedesktop/login1 org.freedesktop.login1.Manager KillUserProcesses (should now show false OR the user should now appear in KillExcludeUsers). Every subsequent step depends on the session no longer being at risk of a mid-work kill.
 2. **Complete RD2-22 live-verify** “ resume the killed regression sweep, curl-verify 401/200, land the Â§11.4.135 guard + HelixQA Challenge entry.
 3. **Resume docs/GOVERNANCE_AUDIT_2026-08-08_ROUND2.md** remaining root-cause-grouped items in priority order; the RD2-40 Â§11.4.238 compliance item is now MECHANICALLY ENFORCED as of this session (INV19-22), but the ledger’s ongoing curation is the standing work.
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CURRENT STATE “ Session 15 (2026-08-08): governance re-audit + P0 security fix + host root-cause + multi-track kickoff

Operator asked for a full deep-research constitutional-compliance audit (rules followed, constitution-derived technology incorporated, gaps/danger-zones/inconsistencies identified, systematic root-cause remediation plan with live verification). Delivered as docs/

GOVERNANCE_AUDIT_2026-08-08_ROUND2.md (Rev 3) “ live re-verification of every item in the 2026-08-07 audit (GA-01..27) + the original 2026-06-14 plan (RW-01..21), a fresh gap scan (RD2-01 through RD2-09), and root-cause forensics on two previously-unattributed “Auto-commit” commits.

Read that document for the full, prioritized, root-cause-grouped remediation plan “ this entry only summarizes what changed THIS session.

What landed (Session 15)

- **Governance audit, Round 2 (docs/ GOVERNANCE_AUDIT_2026-08-08_ROUND2.md).** Confirmed DONE since the 2026-08-07 audit: Python/frontend/extension toolchains (4340 unit tests now collect cleanly), ruff down to 0 real violations (submodule-scoping fix), HelixQA bank symlinks fixed, start.sh --reload-python/--reload-plugins/--recreate real working implementations, CONST-033 triage doc, jackett-autoconfig dead-test removal. Confirmed STILL OPEN: 3 test files mislabeled integration/e2e/contract that mock the system under test, 14 #legacy-untriated tags, Go backend parity (operator-decision, unchanged), stress/chaos coverage gaps, HelixQA hardcoded /Volumes/T7 paths, docker-compose.quality.yml missing container-hygiene limits, a .trivyignore.yml with placeholder CVE IDs, DDoS-class testing fully absent. **NEW P0 finding (RD2-40):** boba does not yet comply with the just-landed constitution anchor Â§11.4.238 (no manual-discovery ledger, no coverage-escape-audit process) “ every finding in this very audit round is itself an instance of the out-of-band-discovery channel that anchor targets.
- **RD2-22 (P0 security) “ source fix DONE, live-verify still owed.** PATCH /api/v1/schedules/{id} and PUT /api/v1/theme were unauthenticated even with BOBA_API_TOKEN set. TDD RED (tests/security/test_hooks_schedules_auth.py extended) “ GREEN (Depends(require_api_token) added to both routes in download-proxy/src/api/scheduler.py + routes.py) “ 31/31 security suite + 176/176 theme/scheduler regression tests pass. A root-cause function-ordering bug (require_api_token referenced before its own definition, crashing the whole API module) was found and fixed during GREEN “ require_api_token moved earlier in routes.py. **Still owed:** the full tests/contract/+tests/unit/ regression sweep (interrupted mid-run by a session kill “ see below) and a live curl-verify (401 unauth / 200 with-token) on a running container.

- **Host root-cause found: KillUserProcesses=true.** The operator reported the session/tmux/ Claude Code process being repeatedly killed. Systematic-debugging (journalctl OOM-killer check â†’ zero hits; systemd-oomd â†’ no kill-decision log lines; live busctl read of org.freedesktop.login1.Manager.KillUserProcesses â†’ confirmed true) found the real mechanism: every time the local GDM/GNOME session closes, systemd-logind SIGKILLS the ENTIRE user@1000.service slice â€” tmux server, Claude Code, everything â€” regardless of Linger=yes. Confirmed 4 occurrences in one day via journalctl, one explicitly listing pid ... (claude) in its kill list.
Awaiting operator decision: a logind.conf.d drop-in (KillUserProcesses=no blanket, or KillExcludeUsers=milosvasic narrower) â€” not yet applied, host-wide change, needs explicit go-ahead.
- **Track 11 â€” multi-track identity permanently adopted.** docs/MULTITRACK.md + scripts/multitrack/track_branch_label.sh (boba-owned wrapper around the constitutionâ€™s shared, inherited-by-reference labeler â€” never edited in place). This checkout is registered as the home of Track 11; trunk (main) work stays T1 per the constitutionâ€™s own hard-coded TRUNK RULE (never overridden); non-trunk work from this checkout labels T11. Execution stays single-track for now per operator instruction â€” no parallel ruler/ subagent streams spawned.
- **New constitution anchor Â§11.4.238** (â€œautomated QA must be the DISCOVERER, not the confirmerâ€” â€” manual QA must find zero new issues). Landed via a **live cross-session collision:** a parallel Claude session (Opus 5, same host observed via a +0500 timezone offset) independently authored and published the same anchor number first; this sessionâ€™s own unpublished duplicate was discarded (never reached any remote, safe to abandon) per Â§11.4.227(B)â€™s anchor-collision rule. constitution submodule pointer bumped + pushed (50c7a66). This ALSO resolved the mystery â€œAuto-commitâ€” pattern from the 2026-08-07 audit â€” it is very likely this same second session/device, not an unknown external actor.
- **workable_items.db reconciled.** BOB-008â€™s DBâ†’i_ZMD body drift fixed (sync md-to-db) plus a genuine Â§11.4.148(D3) finding surfaced and fixed along the way (the Operator-Block-Details UNBLOCK clause needed enumerated [A]/[B] choices, not free prose). BOB-009/BOB-010â€™s evidence_path violations (narrative text instead of a resolvable path) were investigated â€” real evidence identified via git archaeology (commit 0558399; scripts/boba-ctl.sh + scripts/docs_chain.sh (renamed 2026-08-15 BOB-104 to scripts/workable-items-export.sh) still exist and back the claims) â€” but the workable-items tool has **no subcommand to correct a historical item_history.evidence_path** on an already-Completed item; raw SQL was correctly avoided (violates the tool-only-mutation discipline) and a reopen-then-close cycle was correctly avoided (would falsely imply the underlying work was broken). Tracked as an honest, open tooling gap, not silently worked around.

In flight / NEXT ITEM (P0, do this first)

1. **Finish RD2-22:** re-run the full tests/contract/+tests/unit/regression sweep to completion (it was killed mid-run by the KillUserProcesses issue), then live curl-verify 401/200 on a running container, then add the Â§11.4.135 regression guard + HelixQA Challenge entry.
 2. **Operator decision needed:** apply the KillUserProcesses fix (blanket vs.Â narrower “ see above) “ every future session remains at risk of the same abrupt kill until this lands.
 3. Work through docs/GOVERNANCE_AUDIT_2026-08-08_ROUND2.md’s remaining root-cause-grouped items in priority order (Root Causes 3 “6 + the ungrouped P1/P2 items), starting with RD2-40 (Â§11.4.238 compliance) given its zero-grace-period “no escape hatch” clause.
 4. Deferred until the whole backlog above is genuinely clear (operator’s own gating condition): full production release “ install script, user-level systemd/systemctl integration so the stack boots with the system, full docs, then notify the operator it’s ready to use.
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CURRENT STATE “ Session 14 (2026-06-16): search fixed end-to-end + proven on nezha; one P0 in flight

Operator-reported “search is broken / crashing a lot” + “rutracker/nnmclub never authenticate” “ ROOT-CAUSED and FIXED, proven live on the distributed-boot production stack (nezha.local, podman). Definitive end-to-end proof: full-fleet the matrix on nezha returned **2600 results, 23/29 trackers contributing, encoding-crashed: NONE, all four private trackers authenticated** (rutracker 50 / nnmclub 50 / kinozal 50 / iptorrents 49). QA evidence: docs/qa/search-fix-verify-20260616/ (results.md + html/pdf/docx). BUGFIXES.md Rev 19 entries 48 “55.

What landed (Session 14 “ all published at HEAD 646b295)

- **Multi-word query URL-encoding crash, 17 nova3 plugins** (dbd3858 7 maintained + da7d709 10 adopted) “ a literal space crashed urllib (URL can't contain control characters); single-word worked, multi-word silently dropped ~17 plugins “ “search broken / crashing”. Fix: per-plugin percent-encoding. **Verified live** (nezha): scoped the matrix 592, full-fleet 2600, NONE crashed. (BUGFIXES #48)
- **plugin_bad_query_encoding honesty classifier** (33d90f2) “ stopped reporting a self-inflicted encoding bug as a remote plugin_crashed (Â§11.4.6). (BUGFIXES #49)

- **Private-tracker cookie auth** â€” RUTRACKER_COOKIES injection bypasses the CAPTCHA-walled login (2fc29fc) + existing NNMCLUB_COOKIES; cookies via scripts/extract-tracker-cookies.sh (Â§11.4.10). **Verified 50/50** on nezha. (BUGFIXES #50)
- **/auth/status cookie reflection** (9c2f8dc) â€” chips now show cookie-authenticated trackers green before the first search. (BUGFIXES #51)
- **JSON /api/v1/healthz** (137d7ff) â€” was swallowed by the SPA catch-all (returned HTML); now JSON. (BUGFIXES #52)
- **scripts/deploy-remote.sh** â€” **4 silent-failure bugs** (d5b58cb / 9e059d3 / 42cdb02 / e6b9f8f) that kept fixes from landing on the remote (Â§11.4.108): inline-YAML-comment path, rsync abort on container-owned config/, py_compile false â€œSyntax: Invalidâ€, and install-plugin targeting only qbittorrent not qbittorrent-proxy. Pipeline now completes [1/5]â†’[5/5] clean; re-verified via the pipeline (not manual cp). (BUGFIXES #53)
- **kickass Â§11.4.112 Wonâ€™t-fix** (f3e7a4f) â€” 403 behind a Cloudflare/JS challenge, structurally unsolvable by a headless nova3 plugin; documented in docs/research/. (BUGFIXES #54)
- **Regression baseline:** full unit suite at HEAD **4277 passed, 0 failed** (up from 4216 pre-session, zero regression).

Host note

- **Host disk-full** observed this session (T7 / /Volumes/T7/tmp is the temp dir for this work). Keep an eye on free space before any container rebuild or large deploy â€” a full disk silently fails rsync/builds.

In flight â€” NEXT ITEM (P0, do this first)

- **UTF-8 / Cyrillic query crash in 15 plugins (OPEN â€” NOT yet fixed, Â§11.4.6)**. A sibling defect class to #48 surfaced during the #48 discovery extend-pass (Â§11.4.146/Â§11.4.118): the percent-encoding fix addressed the literal-space control-character crash; the non-ASCII (Cyrillic) byte path is a DISTINCT crash. RED reproduction test is in the working tree (untracked): tests/unit/test_plugin_unicode_query_encoding.py. **NEXT:** complete per-plugin UTF-8-safe encoding across the 15 affected engines, flip REDâ†’GREEN (Â§11.4.115), run the Â§11.4.146 extend-pass to enumerate the full affected set, verify live on nezha, then complete BUGFIXES #55 with the GREEN proof + commit hash.

CURRENT STATE â€” Session 13 (2026-06-13): stack booted + qBittorrent 5.x compat

The #1 release blocker is CLEARED. The full multi-container stack is BOOTED and the live detectâ†’sendâ†’torrent round-trip is GREEN â€” the

synthetic infohash is independently confirmed in qBittorrent™'s real torrent list, then cleaned up (§11.4.14).

- **Boot blocker (macOS/podman) €" root-caused + fixed.** / Volumes/T7 (external SSD) was not shared into the podman applehv VM €" podman create hung on an unreachable bind path (proven: py3.13 + 3.14 hung identically; NOT a python-version issue). Fix (operator-authorized §11.4.122): recreated the machine with --volume /Volumes/T7:/Volumes/T7 --cpus 4 --memory 6144; export QBITTORRENT_DATA_DIR=/Volumes/T7/Projects/Boba/downloads; ./start.sh. All 5 containers healthy. Earlier boba-ctl up -d boot bug fixed in 8b6d245.
- **3 REAL product defects (qBittorrent v5.2.1) €" 0dc247b.** The proxy spoke legacy text (200 "Ok.") while linuxserver:latest speaks 5.x JSON/204. download-proxy/src/api/routes.py: _qbit_login_succeeded (204+QBT_SID cookie, all 4 login sites) + _qbit_add_succeeded (modern JSON add + 409-duplicate idempotency, all 3 add sites). Guard: tests/unit/test_qbit_login_compat.py (19 tests). Independent code-review (§11.4.142/§11.4.134): 2 BLOCKING + 1 warning caught €" clean GO. Documented BUGFIXES.md Rev 14 entries 36€"38.
- **macOS LAN access €" ef121e7:** scripts/ensure-macos-tunnel.sh now supports TUNNEL_BIND_ADDR=0.0.0.0 for LAN-IP access (default 127.0.0.1 unchanged).
- **Manual testing (stack UP):** http://localhost:7187/ (dashboard) Â· http://localhost:7186 (qBittorrent admin/admin) Â· :7189 boba-jackett Â· :9117 Jackett. Re-tunnel after reboot: bash scripts/ensure-macos-tunnel.sh (TUNNEL_BIND_ADDR=0.0.0.0 for LAN).
- **Remaining (operator-gated):** headful MV3-load e2e (needs a real display); store assets/submission.
- **Process note:** a background integration suite (test_live_containers.py, which does compose down/up) tore down the live stack mid-session; killed + restored. Parallel work during operator testing MUST stay hermetic (§11.4.119) €" no integration/e2e/container-touching tests.

CURRENT STATE €" Session 10 (BobaLink browser extension)

New major feature in flight: the BobaLink browser extension. Plan + discovery + Phase 1 foundation + backend BE-1/BE-2 + a token-auth security fix landed. The Python unit suite stays FULLY GREEN at 4149 passed.

What is BobaLink

A WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs on any page and forwards them to the running Boba merge service on port **7187**. Backend contract: POST http://

<host>:7187/api/v1/download. The full plan, discovery analysis, and traceability live under docs/browser_extension/ " master plan IMPLEMENTATION_PLAN.md (9 phases), analysis artifacts _analysis/01"06, planning artifacts _plan/A"F, and the 245-item traceability matrix _plan/C-traceability-matrix.md (240 v1 / 5 v2, proving nothing is skipped per Â§11.4.118).

What landed (Session 10 " BobaLink)

- **b2356ae** " BobaLink master implementation plan + discovery/ planning artifacts (docs/browser_extension/); also bumped HelixQA submodule bcac236 " 4d2dcb2.
- **33a9815** " extension **Phase 1 foundation** scaffolded (extension/ " WXT config, TS, shared libs/types under extension/src/); **71 anti-bluff Vitest unit tests** (vitest 71 passed, eslint clean " crypto imports the REAL module, not an inline copy).
- **d46bffb** " backend **BE-1** (CORS for extension origins) + **BE-2** (raw .torrent upload endpoint) added to the Python :7187 API.
- **284d1c4** " bumped HelixQA submodule 4d2dcb2 " bca3b36; new Challenge bank submodules/helixqa/banks/boba-bobalink.yaml (6 cases).
- **192b945** " **security fix**: env-gated shared-secret BOBA_API_TOKEN enforced on the three :7187 download-write endpoints (when the env var is set, requests must carry a matching X-Boba-Token / Bearer token; behaviour unchanged when unset). Verified by **15 token-auth tests** (tests/unit/api_layer/test_download_token_auth.py).

What landed since Session 10 (BobaLink " Phases 2 & 3 + Phase 4 leaf)

- **7225470** " Phase 2 wave-1: parsers (bencode.ts/magnet.ts) + scanner base/site-db.
- **fa03323** " Phase 2 wave-2: .torrent-file SHA-1 infohash + link/text scanners + perf/stress.
- **946c61e** " Phase 2 complete: scanner orchestrator (cross-scanner dedup).
- **2e59572** " lint test files clean; lint script extended to tests/.
- **e8fde43** " Phase 3 shell + Phase 4 api leaf: api/{boba-client,queue} + content/popup/options.
- **15a9a61** " Phase 3 capstone: background service worker (message router). **HEAD.**

Suite status

- Full Python unit suite: **4149 passed** " pytest tests/unit/ --import-mode=importlib (Session 10 baseline; unchanged this session). Evidence: qa-results/tokenauth_fullsuite_*.log.
- Extension Vitest corpus: **379 passed across 33 spec files** " same-session npx vitest run (Tests 379 passed (379)); tsc --noEmit clean; npm run lint 0/0. (+92 over the 287/22 baseline.) Build: npx

wxt build --loadable .output/chrome-mv3/ (11.4.38 -- 8/8 manifest assets verified present).

Current phase + next steps (Session 11 -- two parallel waves landed; reviewed checkpoint)

- **Phases 1, 2, 3 + extension shell + background SW: COMPLETE.**
- **WXT build wiring: COMPLETE** -- endpoints at src/entrypoints/{background.ts,content.ts,popup/index.html,options/index.html} (thin wrappers); npx wxt build --loadable .output/chrome-mv3/; matches derived from SITE_SELECTORS (24 hosts, no <all_urls>); least-privilege manifest; 11.4.38 verified.
- **Phase 4 (backend integration): IN PROGRESS** -- api leaf @e8fde43; **Phase-7 decrypt-before-send WIRED** -- BobaClient.create() decrypts the encryptedBobaApiToken bundle; background reads the session passphrase from chrome.storage.session, sends decrypted plaintext, default-open when locked (RED--GREEN; token/passphrase never logged). **PENDING:** live-7187 integration (require_backend(7187)) + detect--send--torrent-in-qBittorrent E2E on the real backend.
- **Phase 5 (tab groups): IN PROGRESS** -- standalone src/tabgroups/index.ts (dedupe-across-group + batch dispatch) + 13 tests landed; **PENDING:** wire into background MENU_SEND_GROUP + add tabGroups/tabs perms (least-privilege review first).
- **Phase 6 (i18n): IN PROGRESS** -- locale.test.ts guards en-catalog completeness. **Phase 7 (security): IN PROGRESS** -- decrypt path + tests/security/* (least-privilege/CSP/no-hardcoded-secret/secret-storage). **Phase 8: IN PROGRESS** -- 379/33 green; real-module Challenge challenges/extension/detect_and_forward_challenge.sh (mutation-verified); HelixQA BOBA-LINK-007; e2e is an honest operator-gated SKIP (sandbox can't load extensions). **Phase 9: PENDING** -- ci-ext.sh gate + per-store packaging + 11.4.65 doc siblings.
- **Next actions:** (1) **independent-subagent code-review re-pass** for the correctness/security/build lenses once the platform subagent-dispatch throttle clears (done in-context this session; re-run before any release tag per 11.4.40); (2) Phase 5 integration into background + perms; (3) Phase 4 live-7187 integration + E2E; (4) Phase 9 packaging; (5) regenerate Status_Summary.md + 11.4.65 HTML/PDF siblings for the browser_extension Status docs. Status: docs/browser_extension/Status.md (Rev 2, accurate). QA evidence: docs/qa/bobalink-2026-06-10-session11/.

CURRENT STATE -- Session 10 (2026-06-10, subagent-driven loop)

Suite stays FULLY GREEN. This session was hardening + verification, subagent-driven (11.4.70), every change code-reviewed before commit (11.4.125/11.4.134).

What landed (Session 10)

- **test_credential_env_wiring.py isolation fix** “ resolved the Session-9 “known low-priority follow-up” below. A `merge_service` fake-namespace stub was installed at module-collection time with no teardown, so `conftest`’s `_isolate_download_proxy_modules` captured-and-restored the *fake* stub into `siblings` in some run orders. Fixed by moving stub install into a `search_mod` fixture that snapshots/restores the 3 injected keys (`merge_service`, `.search`, `.retry`) in finally. **Evidence:** RED reproduction (the 5 credential tests + a temporary leak-probe) went 1 failed “ all green; the file’s own **5 tests pass** standalone; full unit suite **4121 passed** — seeds default/42/31337 (Session-10 run). Code-review **GO** (proved negation; no weakened assertions; ruff clean).
- **docs/qa/BOB-008/operator_runbook.md** “ copy-pasteable operator unblock procedure for BOB-008 (cookie path preferred; CAPTCHA path documented with its `cap_sid/cap_code` field friction), every endpoint claim cited to `auth.py:line` (Â§11.4.83/Â§11.4.99).
- **Latent-leak discovery sweep (Â§11.4.118)** “ audited all of tests/ for the module-level fake-stub-no-teardown pattern. Finding: the pattern is a **structural no-op wherever the test dir has an `__init__.py`** (real package pre-loads “ `setdefault` never installs the fake), so the ~26 tests/unit/`merge_service/` matches and `test_tracker_stats.py` are **inert** (no RED reproduces; strict TDD “ no fix). The single genuinely-uncovered root is `pirateiro` (see follow-up).
- **Constitution inheritance verified (Â§11.4.32/Â§11.4.35)** “ PASS: pointer present in both layers, propagated anchors present in canonical source, no conflict markers, `pre_build_verification.sh` 18/18 green.
- **pirateiro test-isolation fix** “ **BOB-063 (operator-approved, implemented)** “ tests/unit/test_plugin_pirateiro.py injected `sys.modules['pirateiro']` at module scope with no teardown; `pirateiro` was the one root uncovered by `conftest`’s isolation, so it leaked into later tests. The naive “add to `POLLUTING_ROOTS`” does NOT work (the leak is already inside each per-test snapshot); the real fix caches+re-registers+purges the stub per unit test. **Evidence (Â§11.4.115):** RED 1 failed, 44 passed “ GREEN 45 passed; negation proof (disable the purge “ re-fails); full suite **4122 passed** — seeds default/42. Standing regression guard tests/unit/test_pirateiro_isolation_guard.py (Â§11.4.135). QA at docs/qa/BOB-063/evidence.md. **Closed BOB-063** (Task “ Completed).
- **Constitution submodule advanced 60e2d66 “ f26368b (Â§11.4.26, clean fast-forward, Â§11.4.113-safe)** “ pulled Â§11.4.140 v2 (BACKGROUND action), Â§11.4.142 (universal code-review “ already actively enforced this session; added to CLAUDE.md propagated clauses), Â§11.4.143 (video-streaming real-user-journey “ latent/N-A to this torrent-proxy project). Cascade gate **18/18 green** at the new pin.

Open queue (Issues.md)

- **BOB-008** “RuTracker CAPTCHA” **OPERATOR-BLOCKED**. Unblock procedure documented at docs/qa/BOB-008/operator_runbook.md (preferred: paste bb_session via POST /api/v1/auth/rutacker/cookie-login).
- Everything else closed (DB “i_ZMD in sync, **63 items**, bin/workable-items validate OK).

Known low-priority follow-up

- ~~pirateiro-test-isolation-defense-in-depth~~ “**RESOLVED this session** (BOB-063, see above).
- ~~test_credential_env_wiring/env_loader_stub-teardown-quirk~~ “**RESOLVED this session** (see above).

CURRENT STATE “Session 9 (2026-06-10, overnight autonomous loop)”

The unit suite is now **FULLY GREEN** and **DETERMINISTIC**.

Metric	Value
Unit tests	4121 passed, 0 failed “pytest tests/unit/ --import-mode=importlib (5m24s)
Determinism (Â§11.4.50)	top-level scope 3150 passed identical across --randomly-seed 7/42/100/31337/12345
Commits this session	6e15a8d (crash guards + async/loop hangs), 6230865 (test-pollution + network timeouts) “both pushed to all upstreams
Code review	GO (zero findings/warnings) after Â§11.4.134 iterate-until-GO

What landed (Session 9)

- **8 product fixes** (6e15a8d): degenerate-input crash guards for tokyotoshokan/kickass/yts/piratebay; enricher full-suite-hang fix (aiohttp.ClientTimeout —6); kickass/bitsearch/torrentgalaxy unbounded-loop caps (MAX_PAGES=50); mutation-scanner .venv scope fix; +7 download_proxy tests. Closed **BOB-060**.
- **Test-suite stabilization** (6230865): eliminated all Â§11.4.50 order-dependent pollution “tests/conftest.py _CORRECT_MS_PATH repo-root fix (the dominant bug: corrupted merge_service.__path__ “broke 11 tests incl.Â all scheduler_api), extended POLLUTING_ROOTS, + per-test socket.socket & os.environ snapshot/restore. Network-I/O

timeout hardening (search.py/routes.py/helpers.py/eztv.py). Closed
BOB-061, BOB-062.

Open queue (Issues.md)

- **BOB-008** â€” RuTracker CAPTCHA â€” **OPERATOR-BLOCKED:** needs you to complete the CAPTCHA at /api/v1/auth/rutacker/captcha + /login, OR paste a fresh bb_session cookie via /auth/rutacker/cookie-login. Cannot be solved autonomously.
- Everything else closed (DBâ†”i, ŽMD in sync, 62 items, bin/workable-items validate OK).

How to run the suite (both work now)

- Monolithic: .venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib â†’ 4121 passed.
- Per-scope (faster, parallelizable): tests/unit/merge_service/ (801) Â· tests/unit/api_layer/ (170) Â· tests/unit/*.py top-level (3150).
- Always pass --import-mode=importlib (the merge_service.deduplicator lazy import needs it; a bare per-file run errors).

Known low-priority follow-up (NOT blocking â€” suite is green)

- A residual test_credential_env_wiring / env_loader sys.modules-stub ordering quirk surfaces only in narrow isolated file-combos; it does NOT manifest in the full top-level scope (5 seeds green) because the new env/sys.modules isolation covers it there. Tighten the env_loader stub teardown if it ever resurfaces.

Session 8 Summary

Metric	Value
Test files	47 test modules (222 test_*.py files total across all suites)
Test cases	4,074+ passed
Coverage (unit)	88%+ (gate: 49%)
Bugs fixed	21 (BOB-001 through BOB-025, minus BOB-008)
Submodules aligned	constitution, challenges, containers, helixqa, jaxkett
Ruff / Mypy	pre-existing warnings only, 0 new

Bug Tracker

21 bugs fixed across Session 8: - 10 B-substring parsing fixes - 3 import re missing fixes - 2 comma-size parsing fixes - 2 crash guards - 1 bt4g hang fix -

1 dedup circular import fix - 1 env_loader flaky test fix - 1 pirateiro full test suite (44/44)

1 remaining: BOB-008 (RuTracker CAPTCHA) " operator-blocked, needs manual cookie paste.

Test File Inventory

tests/unit/merge_service/ (31 files)

- test_cors_config.py
- test_dead_tracker_bucket.py
- test_deadline_tunable.py
- test_deduplicator.py
- test_deduplicator_edge.py
- test_diag_no_stale_leakage.py
- test_edge_case_challenges.py
- test_enricher.py
- test_enricher_edge.py
- test_hooks.py
- test_html_parsers.py
- test_jackett_autoconfig.py
- test_nnmclub_session_login.py
- test_private_tracker_html_fixtures.py
- test_public_plugin_harness.py
- test_public_plugin_harness_broad.py
- test_public_tracker_capture.py
- test_public_tracker_subprocess_timeout.py
- test_quality_detection.py
- test_scheduler.py
- test_scheduler_coverage.py
- test_search_concurrency.py
- test_search_coverage.py
- test_search_deep_coverage.py
- test_search_error_paths.py
- test_session_encryption.py
- test_theme_endpoint.py
- test_tracker_stats.py
- test_ttl_caches.py
- test_validator.py
- test_validator_coverage.py

tests/unit/api_layer/ (14 files)

- test_auth_coverage_extra.py
- test_concurrent_writers.py
- test_cors_config.py
- test_hooks_coverage.py
- test_hooks_endpoints.py
- test_hooks_remaining.py
- test_nnmclub_auth_endpoints.py
- test_routes_coverage.py
- test_sse_disconnect.py
- test_sse_token_auth.py

test_theme_state_coverage.py
test_theme_stream.py
test_tracker_stats_sse.py

tests/unit/ (55 files, infrastructure + plugin tests)
test_auth.py, test_auth_coverage.py, test_auth_models.py
test_config.py, test_main.py, test_dashboard.py
test_content_type_refinement.py
test_credential_env_wiring.py
test_download_proxy_coverage.py, test_download_proxy_deep.py
test_download_merged.py
test_env_loader.py
test_freeleech.py
test_graceful_shutdown.py
test_helpers.py
test_log_filter.py
test_merge_trackers.py
test_novaprinter.py
test_private_tracker_search.py
test_public_tracker_subprocess.py
test_retry_policy.py
test_routes.py, test_routes_coverage.py
test_scheduler_api.py
test_sorting_weights.py
test_sse_disconnect.py
test_streaming.py
test_theme_injector.py, test_theme_wiring.py
test_webui_theme_injector.py
test_ui_module.py, test_ui_sorting.py
test_plugin_*.py (39 plugin modules: eztv, piratebay, s
limetorrents, torlock, gamestorrents,
anilibra, torrentgalaxy, yts, rutor,
snowfl, torrentdownload, linuxtracker
nnmclub, rutracker, megapeer, jackett
one337x, extratorrent, torrentfunk, t
therarbg, academictorrents, ali213, y
glotorrents, pctorrent, rockbox, bitr
torrentscsv, xfsusb, yihua, pirateiro,
test_*_guards.py, test_*_deep.py (crash guards + deep coverage variants)
test_ci_infra.py, test_ci_workflows.py
test_community_plugins_compile.py
test_courses_scaffold.py, test_course_scripts_lint.py
test_docs_presence.py
test_frontend_spec_coverage.py
test_install_plugin_json_config.py
test_jackett_integration.py, test_jackett_plugin_pool.py
test_nnmclub_config_selfheal.py, test_nnmclub_plugin_login.py
test_no_runtime_service_skips.py
test_openapi_frozen.py
test_page_title.py, test_footer_restored.py
test_palette_catalog.py, test_palette_catalog_python_mirror.py
test_quality_compose.py, test_quality_detection.py

```

test_readme_landing_page.py
test_runtime_requirements_includes_new_deps.py
test_scan_script_non_interactive.py
test_scanner_configs.py
test_service_fixtures.py
test_shadow_tokens.py
test_socks_udp.py
test_start_sh_copy_plugins_framework.py
test_toolchain_config.py
test_tracker_validator.py
test_website_config.py
test_branding_assets.py
test_architecture_diagrams.py
test_build_releases_non_interactive.py
tests/benchmark/                (3)
tests/chaos/                     (2)
tests/concurrency/              (3)
tests/contract/                 (5)
tests/docs/                     (3)
tests/e2e/                      (7)
tests/integration/              (19)
tests/memory/                   (1)
tests/observability/            (2)
tests/performance/              (1)
tests/property/                 (2)
tests/security/                 (10)
tests/stress/                   (2)

```

Commits (latest 10+)

```

c8254ee fix: 21 test failures, pirateiro 44/44, coverage gate 88%, dedup c
ccd0dc9 fix: env_loader flaky test + coverage baseline 88.14% + minor test
3d1819c docs: finalize Session 8 " Fixed.md Rev 12, 59 items, 18 bugs, 4
f924841 test: wave 8 " btsow/torrentscsv/xfsusb/yihua (170 tests) + bt4g
6773203 test: wave 7 " yourbittorrent/glotorrents/pctorrent/rockbox/bittr
f191401 test: wave 6 " torrentfunk/torrentproject/therarbg/academicorre
1446cc5 test: wave 5 " audiobookbay/one337x/extratorrent (155 tests) + 3
fd543b0 docs: update CONTINUATION.md for Session 8 (4 waves, 23 plugins)
aae4069 test: wave 4 " kinozal/nmclub/rutracker/megapeer/jackett + mega
e3b7acb test: wave 3 " rutor/tokyotoshokan/snowfl/torrentdownload/linuxtr

```

Coverage Snapshot

Module	Coverage
api/init.py	98%
api/auth.py	91%
api/routes.py	95%
api/streaming.py	99%

Module	Coverage
main.py	94%
deduplicator.py	94%
enricher.py	100%
hooks.py	95%
jackett_autoconfig.py	99%
scheduler.py	93%
search.py	~90%
validator.py	92%
theme_injector.py	99%
env_loader.py	100%
download_proxy.py	~95%
TOTAL (unit)	88%+

Known Issues

1. **BOB-008:** RuTracker CAPTCHA “ operator-blocked (needs manual cookie paste)
 2. Go backend is a skeleton (documented in AGENTS.md)
 3. Containers may be down on session start “ bash start.sh first
 4. macOS + podman network_mode: host does NOT forward ports “ ensure-macos-tunnel.sh handles this
-

Quick-Start

```
# One-time venv bootstrap (gitignored .venv/ “ see docs/TESTING.md "Boot
python3 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install -r tests/requirements.txt

# Pre-build gate
bash scripts/pre_build_verification.sh

# Unit tests
.venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib

# Coverage
.venv/bin/python -m pytest tests/unit/ --cov=download-proxy/src --cov=plug

# Lint + typecheck
.venv/bin/ruff check . && .venv/bin/mypy download-proxy/src/

# Frontend
cd frontend && npx vitest run
```

```
# Containers
bash start.sh && bash stop.sh
```

Architecture

Port	Service	Tech
7185	qBittorrent WebUI	LinuxServer (admin/admin)
7186	Download Proxy	Python HTTP
7187	Merge Search Service	FastAPI + Angular SPA
7188	WebUI Bridge	Python (host process)
7189	boba-jackett	Go/Gin
9117	Jackett	LinuxServer (auto-configured)